Practice Set 2 Paper 2 – Urology

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Practice paper compiled from the question bank.

Q1. Discuss evaluation & management of reno-vascular hypertension

Type: LAQ (25 marks) (also answered in 2022-Summer Paper 2, Q2)

Definitions

- **RVH (renovascular hypertension):** hypertension caused by RAS (renal artery stenosis) that improves after revascularisation or nephrectomy. RAS is common in older adults; RVH is proved only by cure.
- **Ischaemic nephropathy:** loss of GFR (glomerular filtration rate) from RAS affecting the whole renal mass (bilateral RAS or stenosis to a solitary kidney).

Causes

	Atherosclerotic RAS (about 90%)	FMD (fibromuscular dysplasia)
Patient	Older, smoker, diabetic, vascular disease	Young women
Site	Ostium and proximal third	Mid and distal artery; "string of beads" (medial fibroplasia)
Course	Progresses; renal function lost	Rarely occludes

Also know: Takayasu arteritis, neurofibromatosis, midaortic syndrome (children), aneurysm, dissection, embolism, Page kidney, transplant RAS.

Pathophysiology (basis of the tests)

- Low perfusion pressure releases **juxtaglomerular renin**; angiotensin II causes vasoconstriction and aldosterone causes sodium retention.
- **Two-kidney, one-clip** (unilateral RAS): renin-dependent; the normal kidney excretes the sodium, so ACE (angiotensin-converting enzyme) inhibitors lower BP (blood pressure) and the captopril test is positive.
- **One-kidney, one-clip** (bilateral or solitary): volume-dependent; ACE inhibitors may cause acute renal failure.
- The stenotic kidney keeps its GFR by angiotensin II **efferent constriction**. Later, hypertension damages the contralateral kidney, so cure after revascularisation falls with duration.

Evaluation

1. Clinical clues (who to test)

- Onset below 30 years (FMD) or severe hypertension after 55 years; **resistant** hypertension (3 or more drugs) or sudden worsening.
- **Creatinine rise above 30% after ACE inhibitor or ARB (angiotensin receptor blocker)**; **flash pulmonary oedema**; kidneys differing by more than 1.5 cm; abdominal bruit; hypokalaemia.

2. Basic tests: creatinine and eGFR (estimated GFR), potassium, urinalysis and proteinuria, ECG; exclude primary aldosteronism and phaeochromocytoma.

3. Imaging and functional tests

Test	Method and finding	Comment
Duplex Doppler US	PSV (peak systolic velocity) above about 180–200 cm/s; renal/aortic velocity ratio >3.5; **RI (resistive index) = (PSV − EDV)/PSV**	First line, no contrast; operator dependent. RI ≥0.8 predicts poor response
CT angiography	Stenosis, ostial plaque, beads	Contrast nephropathy
MR angiography	Gadolinium-enhanced	Caution if eGFR <30 mL/min
Captopril renography (DTPA, diethylenetriaminepentaacetic acid)	Day 1 after oral captopril: slow uptake, low peak; repeat off ACE inhibitor and calcium channel blocker. Improvement of 10% or more = high probability RVH	DTPA is cleared purely by filtration; use declining
Catheter angiography	Gold standard; **translesional gradient** (systolic ≥20 or mean ≥10 mmHg = significant)	Done when intervention is planned
Renal vein renin	Ratio >1.5 lateralises	Before nephrectomy

Algorithm

```
Clinical clue -> Duplex US (or CTA/MRA if eGFR allows)
  -> no RAS: treat as essential hypertension
  -> RAS: FMD -> angioplasty
          Atherosclerotic -> optimal medical therapy
             -> high-risk features + viable kidney -> angiography + gradient -> stent
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Management

1. Medical therapy (first line for atherosclerotic RAS)

- **ACE inhibitor or ARB**, with creatinine and potassium checked at 1–2 weeks; calcium channel blockers, diuretics.
- **Statin, antiplatelet, stop smoking, control diabetes**: most deaths are cardiovascular.

2. Landmark trials

- **ASTRAL (2009):** stenting plus medical therapy gave **no benefit** in renal function, BP, events or death, with procedural complications.
- **CORAL (2014):** stenting did **not reduce** cardiovascular or renal events; small BP fall only.
- Hence no routine revascularisation for atherosclerotic RAS (ESC [European Society of Cardiology] 2017).

3. Indications for revascularisation

- **FMD with hypertension.**
- Atherosclerotic RAS with **flash pulmonary oedema** or recurrent heart failure, **rapidly declining function** in bilateral or solitary-kidney RAS, or truly resistant hypertension.
- Kidney must be viable (not small and atrophic; RI <0.8).

4. Endovascular

- **PTRA (percutaneous transluminal renal angioplasty) alone for FMD**; stent only for dissection or failure.
- **Ostial atherosclerotic lesion: angioplasty plus stent**, because balloon alone fails from elastic recoil of aortic plaque. Restenosis is the main late problem.

- Complications: contrast nephropathy, **atheroembolism**, dissection, rupture, thrombosis, access haematoma.

5. Surgical revascularisation (failed endovascular, branch disease, aneurysm, children, aortic surgery)

Technique	Use
Aortorenal bypass (saphenous vein; internal iliac artery in children; PTFE [polytetrafluoroethylene])	Standard
Transaortic endarterectomy	Bilateral ostial plaque
Splenorenal (left) or hepatorenal (right) bypass	Diseased aorta
Ex vivo bench repair and autotransplantation	Branch FMD, aneurysm
Nephrectomy (partial or total)	Small non-functioning kidney with lateralising renin

6. Transplant RAS: raised velocity on Doppler, contrast imaging confirms; angioplasty or stent first, surgery if this fails.

Follow-up

- BP, creatinine and duplex for restenosis; cure is commoner in FMD than atherosclerosis.

Diagram to draw: Goldblatt two-kidney, one-clip loop (low pressure, renin, angiotensin II, aldosterone, hypertension).

> **Key points to remember:**
> - RVH is proved only by cure; duplex first, angiography with gradient before intervention.
> - ASTRAL 2009 and CORAL 2014: no routine stenting of atherosclerotic RAS.
> - FMD: balloon angioplasty without stent; ostial atheroma: stent.
> - Revascularise for flash pulmonary oedema or declining function in a viable kidney (RI <0.8).

Q2. Management of upper ureteric stones

Type: LAQ (25 marks) (SWL also answered in 2019-Summer Paper 2, Q2 and 2021-Summer Paper 1, Q1; flexible URS in 2019-Summer Paper 2, Q3b)

Introduction

- **Upper (proximal) ureter:** UPJ (ureteropelvic junction) to the upper border of the sacroiliac joint.
- Proximal stones pass least often, so many need intervention. Dangers: renal damage and sepsis.

Evaluation

- **NCCT (non-contrast CT)**: gold standard; gives size, density (HU, Hounsfield units), SSD (skin-to-stone distance) and impaction. KUB (kidney–ureter–bladder radiograph) for SWL follow-up; creatinine, urine culture, coagulation.

1. Emergency: drain first

- **AUA (American Urological Association) indications for urgent intervention:** infection with obstruction, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Obstructed infected kidney: drain only** (double-J stent or PCN, percutaneous nephrostomy; equal for sepsis); definitive treatment later. PCN suits large stones (stent fails if transverse diameter ≥9.5 mm).

2. Observation and MET (medical expulsive therapy): limits

Spontaneous passage	Rate
By size	1–4 mm 78%; 5–7 mm 60%; 8–10 mm 39%
By site	UVJ (ureterovesical junction) 92%; distal 75%; mid 60%; **proximal 48%**

- **Observation** suits small (<5–6 mm) uncomplicated stones: NSAIDs (non-steroidal anti-inflammatory drugs) first.
- **MET (alpha-blockers, tamsulosin or silodosin):** reduce spasm so a larger urine bolus pushes the stone. Benefit is clear only for **distal stones 5–10 mm**; meta-analysis shows **no benefit for proximal stones**. SUSPEND (2015) showed no overall benefit.
- Trial of passage is limited to **4–6 weeks**, with imaging.

3. Choice of active treatment

- Factors: stone (size, density, impaction), patient (habitus, anticoagulation, preference), surgeon and equipment.

EAU (European Association of Urology) recommendation for proximal ureteric stones

Size	First choice	Alternative
<10 mm	SWL or URS (ureteroscopy)	–
>10 mm	URS (retrograde or antegrade)	SWL
Large (>15–20 mm), impacted	PCNL with antegrade URS	Laparoscopic ureterolithotomy

SWL versus URS

	SWL (shock wave lithotripsy)	URS
Stone-free rate (AUA/EAU meta-analysis, approximate)	<10 mm about 90%; >10 mm about 70%	<10 mm about 80%; >10 mm about 80%
Sessions	More retreatment	Single session; higher SFR at 2 weeks, equal at 3 months
Anaesthesia	Sedation or none	Usually general
Complications	Fewer	Perforation, stent pain
Anticoagulation	Must stop	Laser URS can continue

- **SWL fails** with size >10–15 mm, impaction (CT "ring" sign), HU >1000, SSD >10 cm, cystine or COM (calcium oxalate monohydrate) stones.
- **SWL technique:** supine; 60–90 shocks/min; ramp energy; no routine pre-stent. Contraindicated in pregnancy, bleeding disorder, untreated UTI (urinary tract infection), distal obstruction, uncontrolled hypertension.

4. Ureteroscopy for proximal stones

- **Flexible URS** preferred: it follows a stone that migrates into the kidney.
- Safety wire; access sheath (12/14 Fr); tight ureter: pre-stent and return after 2–4 weeks.
- **Laser:** holmium:YAG (2120 nm) or thulium fibre laser; dusting or fragmenting; keep **≤10 W in the ureter** to avoid thermal injury.
- **Retropulsion** into the kidney: reduced by low-energy settings and antiretropulsion devices (Stone Cone, NTrap); pneumatic lithotripsy works poorly above the brim.
- **Push-back:** stone deliberately pushed into the renal pelvis with a catheter, then treated by SWL, RIRS (retrograde intrarenal surgery) or PCNL; now largely replaced by in-situ flexible URS.
- Stent only for perforation, impaction, infection or a planned second stage.

5. Antegrade approach (PCNL with antegrade URS)

- Indications: large (>15–20 mm), hard or **impacted proximal stones**, failed retrograde access, urinary diversion, transplant, concomitant renal stones.
- **Upper or mid-pole posterior calyx** gives a straight line to the UPJ; flexible nephroscope or antegrade URS down the dilated ureter; high SFR.

6. Laparoscopic (or robotic) ureterolithotomy

- Indications: large impacted stones, failed SWL/URS, where endourology is unavailable, or with concomitant UPJ obstruction (pyeloplasty).
- Steps: retroperitoneal or transperitoneal access; identify the bulge; **clamp the ureter above the stone** to prevent migration; longitudinal ureterotomy; remove; double-J stent; close with 4-0 absorbable sutures; drain.
- High single-stage SFR; risk of urine leak. **Open ureterolithotomy** is now rare.

Complications of treatment

Complication	Points
Perforation	0–4%; stop, stent about 4 weeks
Avulsion	0.06–0.5%; **proximal third most at risk**; nephrostomy and delayed reconstruction
Stricture	3–6%; image all after URS
Sepsis	About 5% after URS

Complication	Points
SWL	Steinstrasse, haematoma, colic

Also know: pregnancy: conservative, then stent or URS (no SWL).

Recent advances

- Single-use digital scopes; pressure sensors; thulium fibre laser; robotic URS; **BWL (burst wave lithotripsy)**.

Diagram to draw: algorithm for proximal stone: drain if infected; <10 mm SWL or URS; >10 mm URS; large impacted antegrade or laparoscopic.

Key points to remember:

- Proximal stones pass in only 48%; MET does not help proximal stones.
- <10 mm: SWL equals URS; >10 mm: URS first (EAU).
- Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
- Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

Q3a. Bladder dysfunction non genitourinary systemic disease

Type: SAQ (10 marks)

Taken as lower urinary tract dysfunction caused by systemic (non-genitourinary) diseases.

Principle

- The **site of the lesion**, not the disease name, predicts the bladder:
  - Brain above the pons: **DO (detrusor overactivity), no DSD (detrusor–sphincter dyssynergia)**.
  - Spinal cord: DO **with DSD**.
  - Peripheral nerves or roots: **areflexia, reduced sensation, high PVR (post-void residual)**.
- **Urodynamics** before treatment, because symptoms do not predict findings.

Diabetes mellitus (most important)

- **Diabetic cystopathy:** sensory then motor autonomic neuropathy. Impaired sensation leads to large capacity, poor contractility, low flow and later high PVR and UTI (urinary tract infection).
- **DO is in fact the commonest urodynamic finding** (early disease or coexisting cerebrovascular disease); hypocontractility comes late.
- Mechanisms: glycation end products, sorbitol pathway, oxidative stress, polyuria.
- Differentiate from BOO (bladder outlet obstruction) by pressure–flow study. Treatment: **timed voiding**, CIC (clean intermittent catheterisation), cautious antimuscarinics.

Other systemic diseases

Group	Disease	Usual finding
Neurological	Stroke	Acute retention, then DO; no true DSD
	Parkinson disease	DO; pseudodyssynergia; TURP (transurethral resection of prostate) possible after excluding MSA (multiple system atrophy)
	Multiple sclerosis	DO about 38%, DO with DSD 29%, areflexia 26%; changes over time
Infective	HIV/AIDS	DO, DSD; areflexia with CMV (cytomegalovirus)
	Herpes zoster/simplex, Guillain–Barré	Retention, areflexia; reversible
	Spinal tuberculosis, Lyme, HTLV-1	DO with DSD or areflexia
Autoimmune and connective tissue	SLE (systemic lupus erythematosus)	DO; lupus cystitis
	Systemic sclerosis	Areflexia, low compliance
Infiltrative	Amyloidosis (transthyretin)	Low sensation, underactivity, SUI (stress urinary incontinence)
Nutritional	B12 deficiency, alcohol (thiamine)	Mixed; reversible with replacement
Endocrine	Hyperthyroidism	Poor emptying; resolves with treatment

Also know: myasthenia gravis; schizophrenia; gastroparesis.

Management principles

- Treat the disease; reversible therapy.
- Storage: antimuscarinics (trospium, or mirabegron in elderly brains), botulinum toxin, neuromodulation.
- Emptying: CIC; avoid Credé with DSD.

Diagram to draw: neuraxis with lesion levels and the bladder pattern at each.

> **Key points to remember:**
>
> - Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
> - Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
> - Urodynamics before treating; reversible treatment first.

Q3b. Mycophenolate Mofetil.

Type: SAQ (10 marks) (immunosuppressants also answered in 2021-Summer Paper 1, Q3f and 2020-Summer Paper 4, Q1)

Introduction

- **MMF (mycophenolate mofetil):** an antimetabolite immunosuppressant; morpholinoethyl ester **prodrug** of MPA (mycophenolic acid). With tacrolimus it forms maintenance therapy in **over 90%** of kidney recipients.

Mechanism

- MPA is a reversible, non-competitive inhibitor of **IMPDH (inosine monophosphate dehydrogenase)**, the key enzyme of **de novo guanosine nucleotide synthesis**.
- **Why lymphocyte-selective:** T and B lymphocytes depend on the de novo pathway; other cells use the salvage pathway. MPA also inhibits the type II isoform found in activated lymphocytes.
- Result: arrest of T- and B-cell proliferation and reduced antibody formation.

Pharmacology

Feature	Detail
Dose (kidney)	**1 g twice daily** oral or IV; EC-MPS (enteric-coated mycophenolate sodium) 720 mg twice daily is equivalent
Metabolism	Hydrolysed to MPA; liver glucuronidation to inactive MPAG (MPA glucuronide); **enterohepatic recirculation**
Excretion	MPAG in urine; accumulates in renal failure
Interactions	**Ciclosporin lowers MPA** (blocks enterohepatic recirculation), tacrolimus does not; antacids, cholestyramine reduce absorption; never with azathioprine

Uses

- Kidney transplant maintenance (tacrolimus + MMF ± steroids); other solid organs.
- Lupus nephritis, vasculitis; steroid-refractory immune-related adverse events of checkpoint inhibitors.

Adverse effects

- **GI (gastrointestinal):** diarrhoea, nausea, colitis (commonest; dose split or EC-MPS).
- **Marrow:** leucopenia, anaemia, thrombocytopenia: blood count weekly in month 1.
- **Infection:** CMV, BK virus (ureteric stricture), rarely PML (progressive multifocal leucoencephalopathy).
- **Malignancy:** skin cancer, PTLD (post-transplant lymphoproliferative disorder).
- **Teratogenic:** miscarriage, ear and facial malformations; contraception mandatory; switch to azathioprine before conception.
- Advantage: **not nephrotoxic**; no hypertension, diabetes or dyslipidaemia.

Evidence

- Three mid-1990s randomised trials (US, European, Tricontinental) showed fewer early acute rejections than azathioprine or placebo, so MMF replaced azathioprine.
- **ELITE-Symphony (2007):** low-dose tacrolimus + MMF + daclizumab + steroids gave the lowest rejection and best GFR.

Diagram to draw: purine synthesis pathway showing IMPDH block (de novo) and the salvage pathway spared.

Key points to remember:

- Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
- 1 g twice daily; ciclosporin lowers MPA levels.
- Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

Q3c. Cushing Syndrome

Type: SAQ (10 marks)

Definitions

- **Cushing syndrome:** symptomatic hypercortisolism. **Cushing disease:** due to a pituitary ACTH (adrenocorticotropic hormone)-secreting adenoma.
- **MACS (mild autonomous cortisol secretion):** autonomous cortisol without clinical signs (replaces "subclinical Cushing"); raises cardiovascular, bone and diabetes risk.

Causes

Group	Cause	Share
Exogenous	Glucocorticoids (oral, inhaled, topical)	Commonest overall
ACTH-dependent (80–85% of endogenous)	Pituitary adenoma (Cushing disease)	About 70%
	Ectopic ACTH (bronchial carcinoid, small cell lung cancer, phaeochromocytoma)	10–15%
ACTH-independent	Adrenal adenoma, ACC (adrenocortical carcinoma), AIMAH (macronodular hyperplasia), PPNAD (pigmented nodular disease; Carney complex)	About 10%

Clinical features

- Central obesity, moon face, plethora (90–100%); purple striae, hypertension, hirsutism, ED (70–90%).
- **Discriminating:** proximal weakness, bruising.
- **Urological:** nephrolithiasis in up to 50%; hypogonadotropic hypogonadism (low libido, ED).

Diagnosis

Step 1: exclude exogenous steroids. Step 2: confirm cortisol excess (two tests)

Test	Method	Abnormal
Overnight 1-mg DST (dexamethasone suppression test)	1 mg at 11 pm; 8 am cortisol	>1.8 µg/dL (50 nmol/L)
Late-night salivary cortisol	11 pm, twice	Raised
24-h UFC (urinary free cortisol)	Two collections	Raised; used in pregnancy

Step 3: locate the source

- **Low ACTH:** adrenal; adrenal CT or MRI.
- **High ACTH:** pituitary MRI; **IPSS (inferior petrosal sinus sampling) with CRH (corticotropin-releasing hormone)**: central gradient = pituitary; none = ectopic (chest CT).
- Pitfalls: pseudo-Cushing (depression, alcohol), oestrogens, enzyme inducers.

Management

Cause	Treatment
Cushing disease	Trans-sphenoidal surgery (cure 60–80%); radiotherapy
Adrenal adenoma	**Laparoscopic adrenalectomy**

Cause	Treatment
ACC	Open radical adrenalectomy, mitotane
Ectopic	Resect source; otherwise bilateral adrenalectomy
AIMAH, PPNAD	Bilateral adrenalectomy

- **Bilateral adrenalectomy:** for failed pituitary surgery or life-threatening hypercortisolism. Cost: lifelong replacement, **Nelson syndrome** (8–29%).
- **Medical (bridge):** metyrapone, ketoconazole, etomidate, mitotane, mifepristone.

Perioperative care

- **Contralateral adrenal is suppressed:** adrenal insufficiency in about 99.7% after surgery for overt Cushing.
- **Hydrocortisone 100 mg IV** on the day of surgery, then slow oral taper; recovery takes a median 11 months.

Diagram to draw: hypothalamic–pituitary–adrenal axis with ACTH-dependent and ACTH-independent sites.

> **Key points to remember:**
> - Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
> - ACTH low = adrenal; high = pituitary/ectopic (IPSS).
> - Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

Q3d. Maximum androgen blockade

Type: SAQ (10 marks) (also answered as Combined Androgen Blockade in 2018-Summer Paper 4, Q3f)

Definition and rationale

- **MAB (maximum androgen blockade)**, also CAB (combined) or total androgen blockade: castration (orchidectomy or LHRH [luteinising hormone-releasing hormone] agonist) **plus a non-steroidal antiandrogen**.
- **Rationale (Labrie):** castration removes about 90–95% of testosterone, but **adrenal androgens** (DHEA, dehydroepiandrosterone; androstenedione) are converted to DHT (dihydrotestosterone) in the prostate. The antiandrogen blocks these at the AR (androgen receptor).

Antiandrogens

Drug	Dose	Signature toxicity
Flutamide	250 mg three times daily	Diarrhoea, hepatotoxicity
Nilutamide	300 mg, then 150 mg daily	Delayed dark adaptation, pneumonitis
Bicalutamide	**50 mg daily** (with LHRH agent)	Best tolerated; gynaecomastia
Cyproterone (steroidal)	100 mg two to three times daily	Cardiovascular, hepatic; no CAB benefit

- Class effects: gynaecomastia, hepatotoxicity, pneumonitis.

Evidence

- **Crawford (1989):** leuprolide + flutamide improved survival (partly flare control); **Eisenberger (1998):** orchidectomy + flutamide, **no** benefit.
- **PCTCG (Prostate Cancer Trialists' Collaborative Group) meta-analysis, 2000** (27 trials, about 8,000 men): with non-steroidal antiandrogens, 5-year survival improved by only **about 2–3%**; cyproterone slightly worse.
- Small gain, more cost and toxicity: **AUA (American Urological Association) advises against long-term CAB**.

Current uses

1. **Flare prevention:** an LHRH agonist raises LH (luteinising hormone) up to 10-fold for 10–20 days, risking cord compression, pain or obstruction with bone metastases. Start antiandrogen **with or before** the agonist for **21–28 days**. Antagonists (degarelix, relugolix) or orchidectomy cause no flare.
2. **Antiandrogen withdrawal:** on progression, stopping bicalutamide gives a PSA (prostate-specific antigen) fall >50% in 15–30% (mutant AR turns the drug into an agonist).

Modern replacement in mHSPC (metastatic hormone-sensitive prostate cancer)

- ADT (androgen deprivation therapy) plus abiraterone, enzalutamide, apalutamide, docetaxel, or triplets (PEACE-1, ARASENS).
- **ENZAMET:** enzalutamide beat a CAB control arm for overall survival.

Diagram to draw: testicular and adrenal androgen sources with LHRH agonist and antiandrogen sites of action.

Key points to remember:

- MAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.
- Newer AR pathway inhibitors beat CAB (ENZAMET).

Q3e. Complications of PCNL

Type: SAQ (10 marks) (mini-PCNL complications in 2022-Summer Paper 1, Q1)

Introduction

- PCNL (percutaneous nephrolithotomy) is the standard for renal stones >2 cm. Most complications arise from **access and tract dilatation**.
- Graded by the **Clavien–Dindo** system (I: no intervention; II: drugs or transfusion; III: intervention; IV: ICU [intensive care unit]; V: death).

Incidence (systematic review, Seitz 2012, quoted by EAU [European Association of Urology])

Complication	Rate
Fever	10.8%
Transfusion	7% (nearer 1% with modern and mini tracts)
Thoracic	1.5%
Sepsis	0.5%
Organ injury	0.4%
Angioembolisation	0.4%
Death	0.05%

Main complications

Complication	Cause	Prevention and management
Acute bleeding	Infundibular or segmental vessel injury	Fornix puncture; place sheath, tamponade with balloon or **Kaye catheter**, clamp nephrostomy; transfuse
Delayed bleeding	Pseudoaneurysm, AV (arteriovenous) fistula	**Selective angioembolisation**
Infection, SIRS (systemic inflammatory response syndrome), sepsis	Infected stone, high intrarenal pressure	Sterile urine, culture-specific antibiotics, low-pressure irrigation; stop and drain if pus
Pleural (pneumothorax, hydrothorax)	Supracostal access, highest above 11th rib	Subcostal where possible, puncture in expiration; chest drain
Colon	Retrorenal colon, left lower pole	CT review; extraperitoneal: double-J stent and withdraw nephrostomy into colon (colostomy tube)
Spleen, liver	Upper-pole access, organomegaly	Ultrasound guidance; conservative or surgery
Collecting system	Perforation, extravasation	Stop, drain; later infundibular stenosis
Fluid absorption	Long operation, perforation	**Normal saline**; hyponatraemia, overload

Also know: venous air embolism, DVT (deep vein thrombosis) and pulmonary embolism, neuropraxia and eye injury (prone), residual fragments, loss of renal function (multiple tracts), urinoma, nephrocutaneous fistula, retained hardware.

Risk factors

- Staghorn, multiple tracts, supracostal access, large tract, long operation, CKD (chronic kidney disease), diabetes, positive urine culture, inexperience.

Reducing risk

- **Tranexamic acid** reduces transfusion (Cochrane).
- Ultrasound-guided access, mini-PCNL, ECIRS (endoscopic combined intrarenal surgery).

Diagram to draw: kidney cross-section with posterior calyx, Brödel line, pleura and retrorenal colon.

> **Key points to remember:**
> - Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
> - Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
> - Supracostal access increases pleural injury; colon injury at left lower pole.

Q3f. VHL gene.

Type: SAQ (10 marks)

Introduction

- **VHL (von Hippel–Lindau)** is a **tumour suppressor gene on chromosome 3p25**, identified in 1993; three exons, 213-amino-acid protein.
- Germline mutation causes **VHL disease** (autosomal dominant, about 1 in 36,000). Somatic VHL loss occurs in most sporadic clear cell RCC (renal cell carcinoma); chromosome 3p is altered in over 90%.

Function: the oxygen-sensing pathway

- VHL protein binds elongin B, elongin C, cullin-2 and RBX1 to form an **E3 ubiquitin ligase**.
- **Normoxia:** hydroxylated **HIF (hypoxia-inducible factor) -1α and -2α** are tagged for proteasomal destruction.
- **VHL loss (or hypoxia):** HIF, chiefly HIF-2α, accumulates, joins ARNT (aryl hydrocarbon receptor nuclear translocator) and switches on **VEGF (vascular endothelial growth factor)**, PDGF (platelet-derived growth factor), TGF-α (transforming growth factor-α), GLUT1, erythropoietin and CA-IX (carbonic anhydrase IX).
- Hence the **hypervascularity** of clear cell RCC.
- **Knudson two-hit:** one germline plus one somatic hit, so VHL tumours are early, bilateral and multifocal.

VHL disease manifestations

- **Clear cell RCC** 25–75% (mean age 39; commonest cause of death); renal and pancreatic cysts.
- **Craniospinal haemangioblastoma** 60–80%; retinal angioma 25–60%.
- **Phaeochromocytoma** 10–25%; pancreatic neuroendocrine tumours; endolymphatic sac tumour; epididymal cystadenoma.

Genotype–phenotype

Type	Mutation	Phaeochromocytoma	RCC
1	Deletions, truncations	Low	High
2A	Missense	High	Low
2B	Missense	High	High
2C	Missense	Phaeochromocytoma only	Absent

Clinical implications

- **Surveillance (VHL Alliance):** eye check yearly from infancy; metanephrines from 5 years; CNS MRI from 11; abdominal MRI every 2 years from 15.
- **3 cm rule:** observe solid renal tumours <3 cm; then nephron-sparing surgery (enucleation) or ablation, after excluding phaeochromocytoma.
- **Targeted therapy:** VEGF TKIs (tyrosine kinase inhibitors), bevacizumab, mTOR (mammalian target of rapamycin) inhibitors.
- **Belzutifan** (HIF-2α inhibitor): FDA (Food and Drug Administration) approved 2021 for VHL disease; renal response 30–50%; anaemia.
- **Also know:** PBRM1, BAP1, SETD2 on 3p.

Diagram to draw: VHL–HIF pathway in normoxia versus VHL loss, with drug targets.

Key points to remember:

- 3p25 tumour suppressor; VHL protein degrades HIF.
- VHL loss: HIF-2α drives VEGF, giving vascular clear cell RCC.
- Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing.
- Belzutifan (HIF-2α inhibitor) approved 2021.

Rapid Revision Sheet – Practice Set 2 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss evaluation of & management of reno-vascular hypertension (LAQ)

- RVH is proved only by cure; duplex first, angiography with gradient before intervention.
- ASTRAL 2009 and CORAL 2014: no routine stenting of atherosclerotic RAS.
- FMD: balloon angioplasty without stent; ostial atheroma: stent.
- Revascularise for flash pulmonary oedema or declining function in a viable kidney (RI <0.8).

Q2. Management of upper ureteric stones (LAQ)

- Proximal stones pass in only 48%; MET does not help proximal stones.
- <10 mm: SWL equals URS; >10 mm: URS first (EAU).
- Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
- Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

Q3a. Bladder dysfunction non genitourinary systemic disease (SAQ)

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

Q3b. Mycophenolate Mofetil. (SAQ)

- Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
- 1 g twice daily; ciclosporin lowers MPA levels.
- Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

Q3c. Cushing Syndrome (SAQ)

- Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
- ACTH low = adrenal; high = pituitary/ectopic (IPSS).
- Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

Q3d. Maximum androgen blockade (SAQ)

- MAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.
- Newer AR pathway inhibitors beat CAB (ENZAMET).

Q3e. Complications of PCNL (SAQ)

- Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
- Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
- Supracostal access increases pleural injury; colon injury at left lower pole.

Q3f. VHL gene. (SAQ)

- 3p25 tumour suppressor; VHL protein degrades HIF.
- VHL loss: HIF-2α drives VEGF, giving vascular clear cell RCC.
- Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing.
- Belzutifan (HIF-2α inhibitor) approved 2021.